Bootstrapping String Scattering off D-branes in AdS

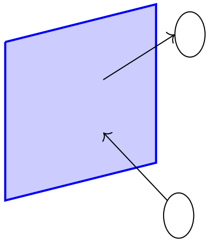
CLÉMENT VIRALLY
(based on work with L. F. Alday and X. Zhou)

Mathematical Institute
University of Oxford

CASE 2026

Introduction: string scattering from D-branes in AdS

Without quantizing the string, can we compute string scattering form factors/amplitudes in AdS?



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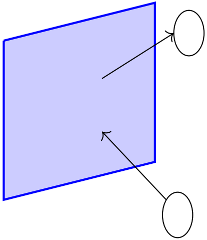

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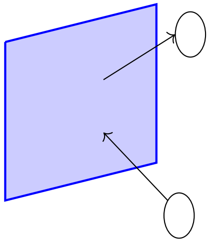

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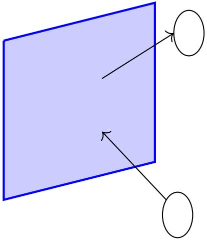

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In the flat space limit, this becomes a tractable bootstrap problem ($\alpha'=1$):

$$\mathcal{G}(S,Q;x)=\underbrace{\mathcal{G}^{(0)}(S,Q;x)}_{\text{known}}+\frac{1}{R^2}\underbrace{\mathcal{G}^{(1)}(S,Q;x)}_{\text{this talk!}}+\dots$$

A bootstrap problem

Even if we can't compute directly, what do we know?

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5. Ansatz for the worldsheet. (Alday, Hansen, Silva 2023; Alday, Chester, Hansen, Zhong 2024)

Structure of the talk

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1. The CFT Side: line defects in $\mathcal{N} = 4$ SYM
2. The AdS side: a worldsheet perspective
3. AdS/CFT: matching and results
4. Outlook

Line defects in $\mathcal{N} = 4$ SYM

- 2-point function of stress tensor multiplet's scalar superprimary $\mathcal{O}(x, u) \equiv \mathcal{O}$ in presence of line defect:

$$\langle\!\langle \mathcal{O}_1 \mathcal{O}_2 \rangle\!\rangle = \frac{(u_1 \cdot \theta)^2 (u_2 \cdot \theta)^2}{|x_1^i|^2 |x_2^i|} \mathcal{F}(\xi, \chi; \sigma)$$

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- Work in Mellin space:

$$\mathcal{F}(\xi, \chi; \sigma) = \int \frac{\mathrm{d}\gamma\, \mathrm{d}\delta}{(2\pi i)^2} \xi^{-\delta} \chi^{-\gamma+\delta} \mathcal{M}(\delta, \gamma; \sigma) \Gamma(\delta) \Gamma(\gamma - \delta) \prod_{i=1}^{2} \Gamma\left(\frac{\Delta_i - \gamma}{2}\right)$$

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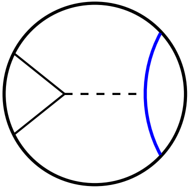

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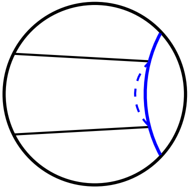
- 2 channels for $\mathcal{M}$:

$$\mathcal{M}^{\text{bulk}}_{\Delta,\ell}(\delta, \gamma) = \sum_{m=0}^{\infty} \frac{R_{\Delta,\ell,m}(\gamma)}{\delta - \frac{\Delta_1 + \Delta_2 - \tau - 2m}{2}}, \quad \mathcal{M}^{\text{def}}_{\Delta,\ell}(\delta, \gamma) = \frac{P_{\Delta,\ell,m}(\delta, \gamma)}{\gamma - \tau - 2m}$$



Superconformal Ward identities

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- $\mathcal{F}$ and thus $\mathcal{M}$ are quadratic in σ:

$$\mathcal{M}(\delta, \gamma; \sigma) = \mathcal{M}_0(\delta, \gamma) + \mathcal{M}_1(\delta, \gamma)\sigma + \mathcal{M}_2(\delta, \gamma)\sigma^2$$

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- Obey superconformal Ward identity:

$$\left(\partial_z + \frac{1}{2}\partial_\alpha\right) \mathcal{F}(z, \bar{z}; \alpha)\Big|_{\alpha = z} = 0, \quad \left(\zeta = \frac{1 + z\bar{z}}{2\sqrt{z\bar{z}}},\ \eta = \frac{z + \bar{z}}{2\sqrt{z\bar{z}}},\ \sigma = -\frac{(1-\alpha)^2}{2\alpha}\right)$$

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- In Mellin space, simple solution ("reduced amplitude" $\widetilde{\mathcal{M}}$):

$$\mathcal{M}(\delta,\gamma;\sigma)=\underbrace{\mathcal{D}(\delta,\gamma;\sigma)}_{\text{shifts}}\quad\underbrace{\widetilde{\mathcal{M}}(\delta,\gamma)}_{\text{no }\sigma\text{, no constraints}}$$

$\mathcal{D}$ can be found universally (in many SCFTs!) by working in a basis that diagonalizes shifts. (CV 2025)

Flat space limit

- Mellin space provides simple flat space limit: (Alday, Zhou 2024)

$$\mathcal{M}_{\Delta,\ell}(\delta,\gamma;\sigma) = \int_0^\infty \mathrm{d}\beta\, e^{-\beta}\sqrt{\beta}\mathcal{A}\left(S = -\frac{2\delta\beta}{R^2}, Q = \frac{2\gamma\beta}{R^2}\right), \quad R \to \infty$$

where $\Delta^{\text{bulk}} = 2\sqrt{n}R + \ldots$, $\Delta^{\text{def}} = \sqrt{n}R + \ldots$ and $m \sim R^2 \implies$ sum over descendants $\to$ integral.

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- Structure:

$$\mathcal{A}^{\text{bulk}}(S,Q)=\frac{P^{\ell/2}_{\text{bulk},n,\ell}(Q)}{S-2n}+\frac{1}{R^2}\left(\frac{\widetilde{P}^{\ell/2}_{\text{bulk},n,\ell}(Q)}{(S-2n)^4}+\text{lower poles}\right)+\dots$$

$$\mathcal{A}^{\text{def}}(S,Q)=\frac{P^{\ell}_{\text{def},n,\ell}(S)}{Q-n}+\frac{1}{R^2}\left(\frac{\widetilde{P}^{\ell}_{\text{def},n,\ell}(S)}{(Q-n)^4}+\text{lower poles}\right)+\dots$$

The worldsheet ansatz

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- Assume a worldsheet form:

$$\mathcal{A}(S,Q)=\int_0^1 \mathrm{d}x\, x^{-\frac{S}{2}-1}(1-x)^{-Q-1}\left(\mathcal{G}^{(0)}(S,Q;x)+\frac{1}{R^2}\mathcal{G}^{(1)}(S,Q;x)+\ldots\right)$$

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- Integrand built from multiple polylogarithms:

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$$\mathcal{G}^{(1)}(S,Q;x) = \sum_{|w|\leq 3} \frac{P^{|w|+1}(S,Q;x)}{(S+2Q)^2} L_w(x) + \zeta, \quad \frac{\mathrm{d}}{\mathrm{d}x} L_{aw}(x) = \frac{1}{x-a} L_w(x)$$

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- Order of polylogarithms to match number of poles from Mellin space.

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- Independent check: agrees with localization results for higher derivative corrections to SUGRA.

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- Future directions: higher orders in $\frac{1}{R^2}$, more defect systems, different dimensions.